

$$1) \quad W(s) = \frac{2}{s(s+4)} \quad \text{PD regulator}$$

$$r(t) = h(t)$$

$$G_{reg}(s) = k_p + k_d s$$

$$W_p(s) = W(s) \cdot G_{reg}(s) = \frac{2k_d s + 2k_p}{s(s+4)}$$

$$f(s) = s^2 + (2k_d + 4)s + 2k_p$$

$$f_z(s) = (s+3)^2 = s^2 + 6s + 9$$

$$2k_d + 4 = 6 \Rightarrow \boxed{k_d = 1}$$

$$2k_p = 9 \Rightarrow \boxed{k_p = \frac{9}{2}}$$

← obratiti pažnju da se polovi postavre tako da se ne dobiju negativne vrednosti parametara

$$2) \quad W(s) = 100 \frac{s+40}{s(s+10)^2(s+80)}$$

$$W(j\omega) = 100 \frac{j\omega + 40}{j\omega (j\omega + 10)^2 (j\omega + 80)}$$

$$W(j\omega) = \frac{100 \cdot 40}{100 \cdot 80} \frac{(j \frac{\omega}{40} + 1)}{j\omega (j \frac{\omega}{10} + 1)^2 (j \frac{\omega}{80} + 1)}$$

$$= \left(\frac{1}{2} \right) \frac{(j \frac{\omega}{40} + 1)}{j\omega (j \frac{\omega}{10} + 1)^2 (j \frac{\omega}{80} + 1)}$$

$$|W(j\omega)|_{dB} = 20 \log \frac{1}{2} + 20 \log \sqrt{\left(\frac{\omega}{40}\right)^2 + 1} - 20 \log \omega - 40 \log \sqrt{\left(\frac{\omega}{10}\right)^2 + 1} - 20 \log \sqrt{\left(\frac{\omega}{80}\right)^2 + 1}$$

prelomne učestanosti: $\omega_1 = 0 \text{ rad/s}$, $\omega_2 = 10 \text{ rad/s}$, $\omega_3 = 40 \text{ rad/s}$,
 $\omega_4 = 80 \text{ rad/s}$

I segment: $\omega \leq 10 \text{ rad/s}$

$$|W(j\omega)|_{dB} = 20 \log \frac{1}{2} - 20 \log \omega$$

$$|W(0.1j)|_{dB} = -6 - 20 \log(0.1) = -6 - (-20) = \underline{\underline{14 \text{ dB}}}$$

$$|W(10j)|_{dB} = -6 - 20 \log 10 = -6 - 20 = \underline{\underline{-26 \text{ dB}}}$$

II segment: $10 \leq \omega \leq 40 \text{ rad/s}$

$$|W(j\omega)|_{dB} = 20 \log \frac{1}{2} - 20 \log \omega - 40 \log \left(\frac{\omega}{10} \right)$$

$$|W(10j)|_{dB} = -6 - 20 \log 10 - 40 \log 1 = -6 - 20 = \underline{\underline{-26 \text{ dB}}}$$

$$|W(40j)|_{dB} = -6 - 20 \log 40 - 40 \log 4 = -6 - 32 - 24 = \underline{\underline{-62 \text{ dB}}}$$

III segment: $40 \leq \omega \leq 80 \text{ rad/s}$

$$|W(j\omega)|_{dB} = 20 \log \frac{1}{2} - 20 \log \omega - 40 \log \left(\frac{\omega}{10} \right) + 20 \log \left(\frac{\omega}{40} \right)$$

$$|W(40j)|_{dB} = -6 - 20 \log 40 - 40 \log 4 + 20 \log 1 = \underline{\underline{-62 \text{ dB}}}$$

$$|W(80j)|_{dB} = -6 - 20 \log 80 - 40 \log 8 + 20 \log 2 = \\ -6 - 38 - 36 + 6 = \underline{\underline{-74 \text{ dB}}}$$

IV segment: $\omega \geq 80 \text{ rad/s}$

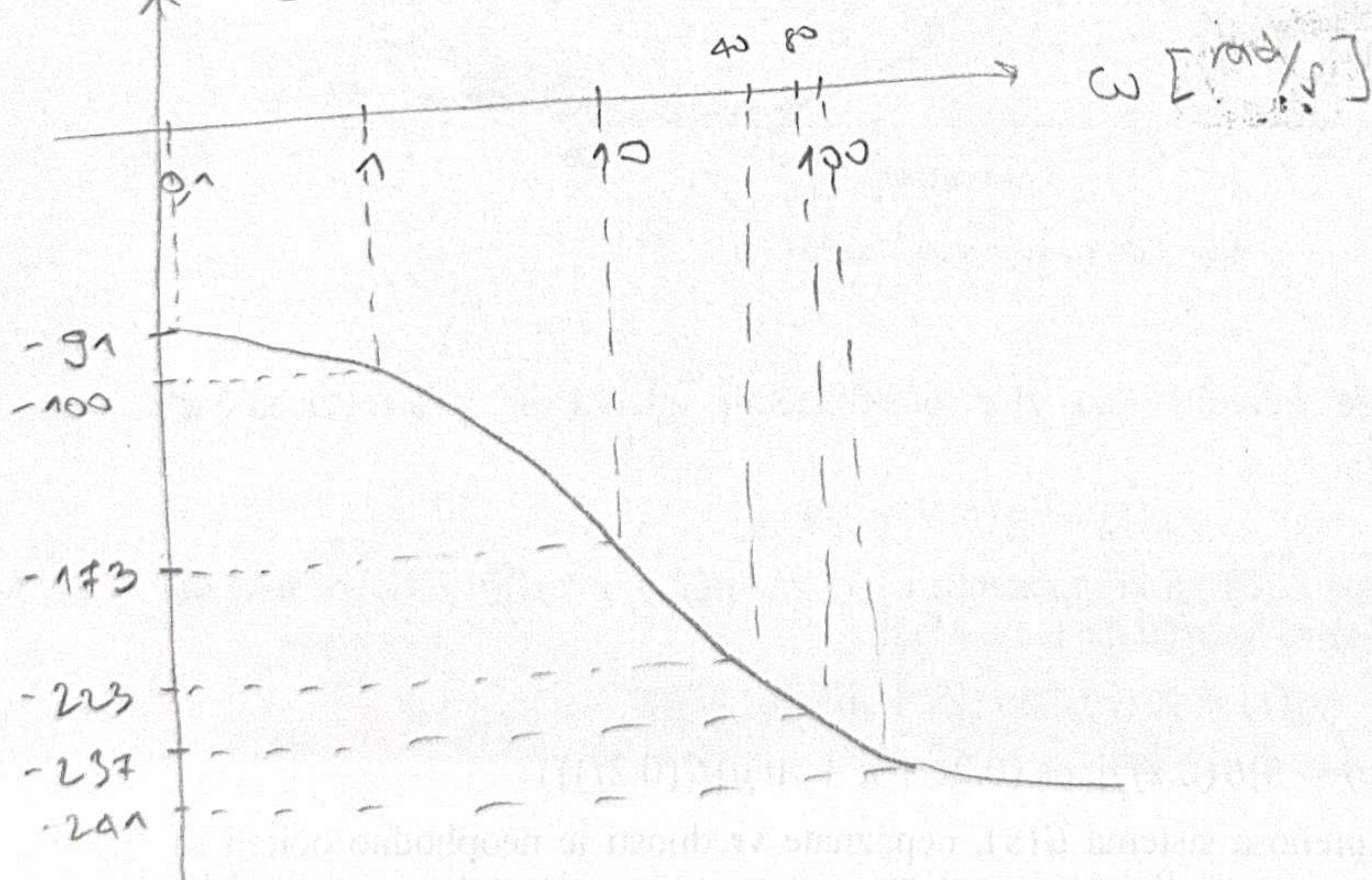
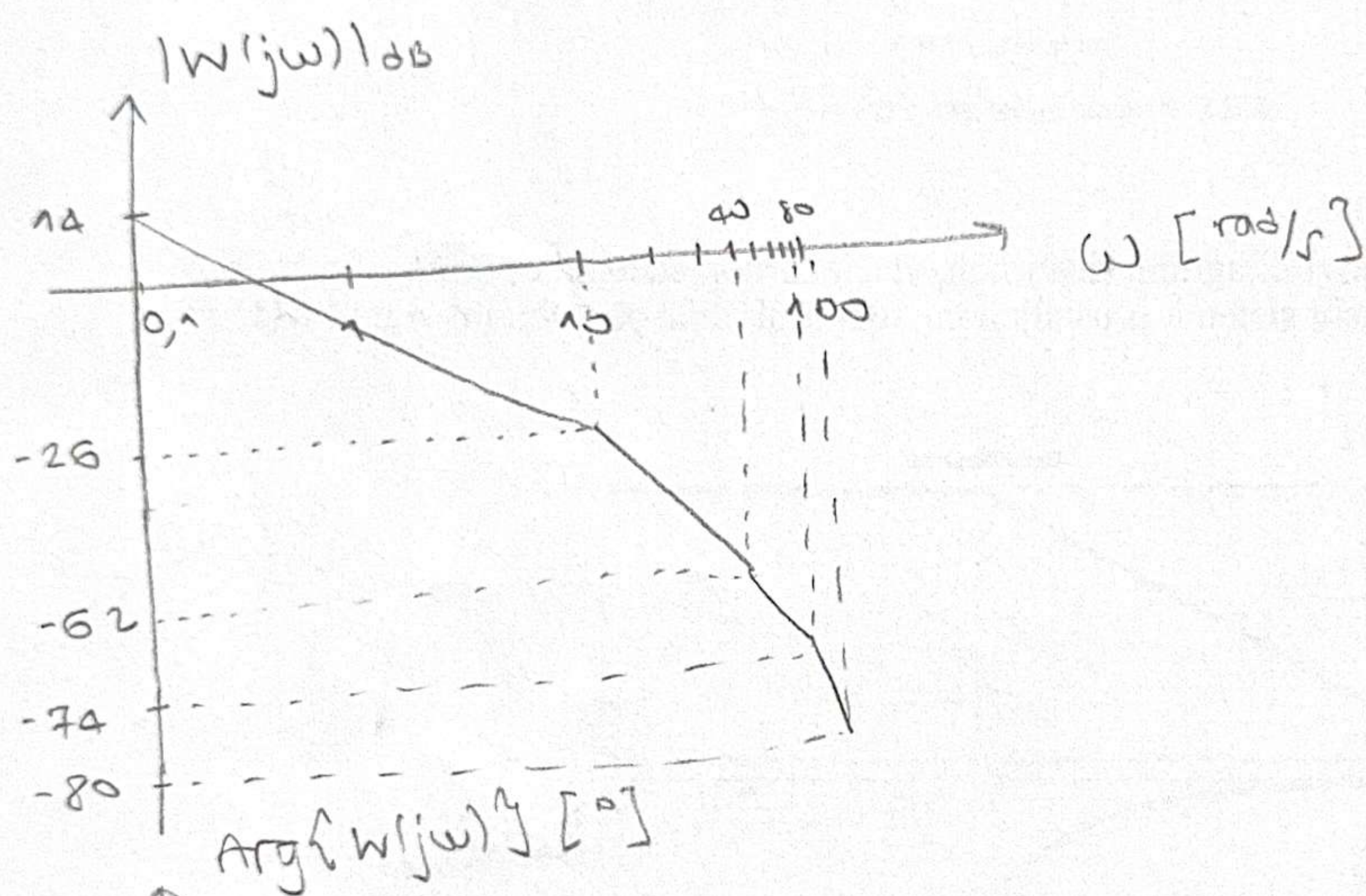
$$|W(j\omega)|_{dB} = 20 \log \frac{1}{2} - 20 \log \omega - 40 \log \left(\frac{\omega}{10} \right) + 20 \log \left(\frac{\omega}{40} \right) - 20 \log \left(\frac{\omega}{80} \right)$$

$$|W(80j)|_{dB} = -6 - 20 \log 80 - 40 \log 8 + 20 \log 2 - 20 \log 1 = \underline{\underline{-74 \text{ dB}}}$$

$$|W(100j)|_{dB} = -6 - 20 \log 100 - 40 \log 10 + 20 \log \left(\frac{100}{40} \right) - 20 \log \left(\frac{100}{80} \right) \\ = -6 - 40 - 40 + 8 - 2 = \underline{\underline{-80 \text{ dB}}}$$

$$\text{Arg}\{W(j\omega)\} = \cancel{\arctg 0} + \arctg \frac{\omega}{40} - \frac{\pi}{2} - 2\arctg \frac{\omega}{10} - \arctg \frac{\omega}{80}$$

ω	$\text{Arg}\{W(j\omega)\} [^\circ]$
0,1	-91
10	-173
40	-223,5
80	-237,3
100	-241,7
1	-100,7



- 3) urađen u PDF-u
- 4) urađen u PDF-u
- 5,8) GMK ne dolazi na test

$$6) W_p(s) = K \frac{(s+10)^2}{s(s+1)(s-50)}$$

$$p=1$$

$$W_{p1}(s) = \frac{W_p(s)}{K}$$

1. početak krive

$$s = \rho e^{j\varphi}, \quad \rho \rightarrow 0, \quad \varphi \in [0, \frac{\pi}{2}]$$

$$W_{p1}(\rho e^{j\varphi}) = \frac{(\rho e^{j\varphi} + 10)^2}{\rho e^{j\varphi} (\rho e^{j\varphi} + 1) (\rho e^{j\varphi} - 50)}$$

$$\lim_{\rho \rightarrow 0} W_{p1}(\rho e^{j\varphi}) = \lim_{\rho \rightarrow 0} \frac{100}{\rho e^{j\varphi} \cdot 50 e^{j\varphi}} = \lim_{\rho \rightarrow 0} \frac{2}{\rho} e^{j(-\varphi - \pi)}$$

$$\lim_{\rho \rightarrow 0} |W_{p1}(\rho e^{j\varphi})| = \infty$$

$$\lim_{\rho \rightarrow 0} \text{Arg} \{W_{p1}(\rho e^{j\varphi})\} = -\varphi - \pi \in [-\pi, -\frac{3\pi}{2}]$$

2. završetak krive

$$s = j\omega, \quad \omega \rightarrow \infty$$

$$W_{p1}(j\omega) = \frac{(j\omega + 10)^2}{j\omega (j\omega + 1) (j\omega - 50)} \cdot \frac{(-j\omega + 10)^2}{-j\omega (-j\omega + 1) (-j\omega - 50)}$$

$$= \frac{3900 - 69\omega^2}{\omega^4 + 2501\omega^2 + 2500} + j \frac{-\omega^4 + 1030\omega^2 + 5000}{\omega^5 + 2501\omega^3 + 2500\omega}$$

$$\omega \rightarrow \infty \quad \begin{cases} \text{Re} \{W(j\omega)\} = 0 \\ \text{Im} \{W(j\omega)\} = 0 \end{cases} \quad \left\{ \begin{array}{l} \lim_{\omega \rightarrow \infty} |W(j\omega)| = 0 \\ 2 \cdot \frac{\pi}{2} = \pi \end{array} \right.$$

$$\lim_{\omega \rightarrow \infty} \text{Arg} \{W(j\omega)\} = \lim_{\omega \rightarrow \infty} (2 \arctg \frac{\omega}{10} - \frac{\pi}{2} - \arctg \omega - (\pi - \arctg \frac{\omega}{50}))$$

$$= \pi - \frac{\pi}{2} - \frac{\pi}{2} - \pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

3. presek sa Re-osom

$$\operatorname{Im}\{w(j\omega)\} = 0$$

$$-w^4 + 1030w^2 + 5000 = 0$$

$$w^4 - 1030w^2 - 5000 = 0$$

$$w^2 = x$$

$$x^2 - 1030x - 5000 = 0$$

$$x_1 = 1034,93$$

$$x_2 = -4,93$$

$$w^2 = 1034,93$$

$$\omega = \underline{\underline{32,17 \text{ rad/s}}}$$

$$\operatorname{Re}\{w(32,17j)\} = \underline{\underline{-0,0184}}$$

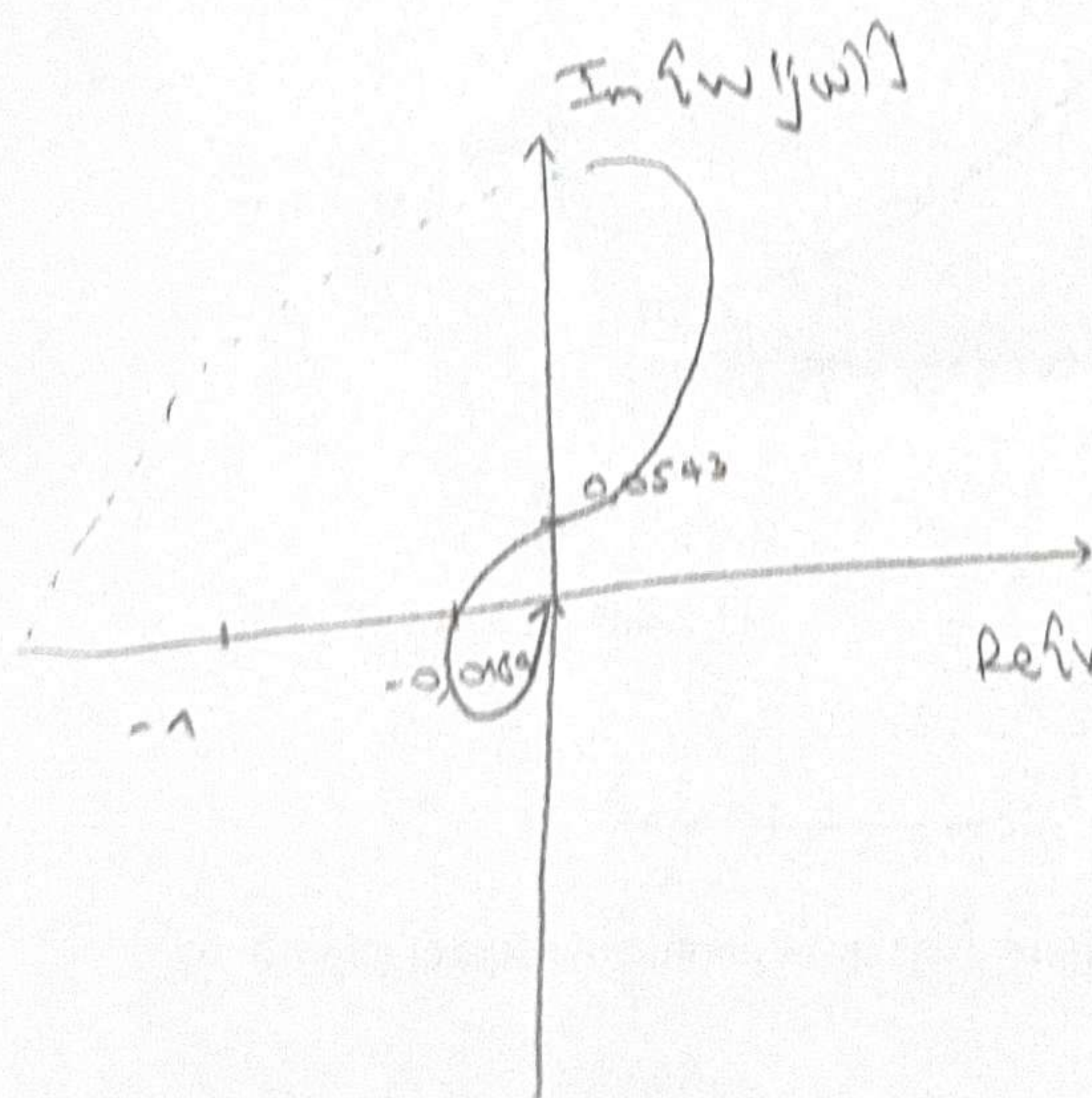
4. presek sa Im-osom

$$\operatorname{Re}\{w(j\omega)\} = 0$$

$$3900 - 69w^2 = 0$$

$$\omega = \underline{\underline{7,52 \text{ rad/s}}}$$

$$\operatorname{Im}\{w(7,52j)\} = \underline{\underline{0,0543}}$$



$$p=1$$

potrebno $\frac{1}{2}$ obuhvata

1. stabilan K.T. $\in (-0,0184, 0)$

2. gr. stab. K.T. $= -0,0184$

3. nestabilan K.T. $\in (-\infty, -0,0184)$

$$1. -\frac{1}{K} > -0,0184$$

$$K > 54,35$$

$$2. K = 54,35$$

$$3. K < 54,35$$

$$7) \quad \ddot{y} + 5\dot{y} + 6y = \dot{u} + 4 \quad / L$$

$$Y(s) (s^2 + 5s + 6) = U(s) (s + 1)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{s+1}{s^2 + 5s + 6} = \frac{s+1}{(s+2)(s+3)} \rightarrow G(j\omega) = \frac{j\omega+1}{(j\omega+2)(j\omega+3)}$$

$$a) \quad u(t) = 3 \sin(t+2)$$

$$y(t) = 3 |G(j1)| \sin(t+2 + \text{Arg}\{G(j1)\})$$

$$|G(j\omega)| = \frac{\sqrt{\omega^2+1}}{\sqrt{\omega^2+2} \sqrt{\omega^2+3}}$$

$$\text{Arg}\{G(j\omega)\} = \arctg \omega - \arctg \frac{\omega}{2} - \arctg \frac{\omega}{3}$$

$$|G(j1)| = \frac{\sqrt{2}}{\sqrt{3} \sqrt{4}} = \frac{1,41}{1,73 \cdot 2} = 0,4$$

$$\text{Arg}\{G(j1)\} = \arctg 1 - \arctg \frac{1}{2} - \arctg \frac{1}{3} = 0,78 - 0,46 - 0,32 = 0$$

$$\underline{\underline{y(t) = 1,2 \sin(t+2)}}$$

$$b) \quad u(t) = \sin(t+1) + 4 \cos(2t+3)$$

$$y(t) = |G(j1)| \sin(t+1 + \text{Arg}\{G(j1)\}) + 4 |G(j2)| \cos(2t+3 + \text{Arg}\{G(j2)\})$$

$$|G(j2)| = \frac{\sqrt{5}}{\sqrt{6} \sqrt{7}} = \frac{2,236}{2,45 \cdot 2,65} = 0,345$$

$$\text{Arg}\{G(j2)\} = \arctg 2 - \arctg 1 - \arctg \frac{2}{3} = 1,1 - 0,78 - 0,588 = -0,268$$

$$\underline{\underline{y(t) = 0,4 \sin(t+1) + 1,38 \cos(2t + 2,732)}}$$

$$g) \quad G(s) = \frac{(s-1)(s+a)}{(s+b)(s-c)(s+2)}$$

$$G(s) = \frac{y(s)}{u(s)} = \frac{s^2 + s(a-1) - a}{s^3 + s^2(2-c+b) + s(-2c+2b-bc) - 2bc} \cdot z(s)$$

$$y(s) = (s^2 + s(a-1) - a) z(s)$$

$$u(s) = (s^3 + s^2(2-c+b) + s(-2c+2b-bc) - 2bc) z(s)$$

$$(s+b)(s^2+2s-cs-2c) = s^3 + 2s^2 - cs^2 - 2cs + bs^2 + 2bs - bcs - 2bc$$

$$= s^3 + s^2(2-c+b) + s(-2c+2b-bc) - 2bc$$

$$x_1(s) = z(s)$$

$$x_2(s) = s z(s) = s x_1(s) \quad / \mathcal{L}^{-1}$$

$$x_2(t) = \dot{x}_1(t)$$

$$x_3(s) = s^2 z(s) = s x_2(s) \quad / \mathcal{L}^{-1}$$

$$x_3(t) = \dot{x}_2(t)$$

$$s x_3(s) = s^3 z(s) = - (2-c+b) x_3(s) - (-2c+2b-bc) x_2(s) + 2bc x_1(s) \quad + u(s)$$

$$\dot{x}_3(t) = (-2+c-b) x_3(t) + (2c-2b+bc) x_2(t) + 2bc x_1(t) + u(t)$$

$$y(s) = x_3(s) + (a-1) x_2(s) - a x_1(s) \quad / \mathcal{L}^{-1}$$

$$y(t) = -a x_1(t) + (a-1) x_2(t) + x_3(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2bc & 2c-2b+bc & -2+c-b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} -a & (a-1) & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$